

SC0350-270-RSS



APPLICATIONS

- Wind Turbine Pitch Control
- Industrial Backup Power
- Electric Power Tools
- Renewable Energy Systems
- Energy Harvesting
- AGV's



FEATURES & ADVANTAGES

- One Million Cycle Life
- Good Low Temperature Characteristics
- Ultra High Power Density
- Ultra Low Internal Resistance
- 10-15 year calendar life



Specifications

Capacitance	Rated ¹	350F
	Tolerance	0/+20%
Voltage	Rated	2.7V DC
	Surge ²	2.85V DC
ESR	ESR (DC) - typical	2.3mΩ
	ESR (DC) - maximum initial	2.8mΩ
Current	Maximum leakage ³	0.3mA
	Maximum Peak	220A
	Maximum continuous current ($\Delta T = 15^{\circ}\text{C}$) ⁴	21A RMS
	Maximum continuous current ($\Delta T = 40^{\circ}\text{C}$) ⁴	34A RMS
Energy Storage	Maximum energy ⁵	0.35Wh
	Usable energy ⁶	0.27Wh
	Volumetric energy density ⁷	6.53Wh/L
	Gravimetric energy density ⁸	5.56Wh/kg
Power Density	Power density ⁹	4967W/kg

Temperature

Temperature Characteristics	Operating Temperature Range ¹⁰	-50°C to +65°C
	Storage Temperature Range	-50°C to +70°C

Safety

Safety	Short Circuit Current	960A
	• This product may vent or rupture if overcharged, reverse charged, incinerated or heated above 100°C • Do not crush, mutilate, or disassemble • Do not dispose of unit in trash	

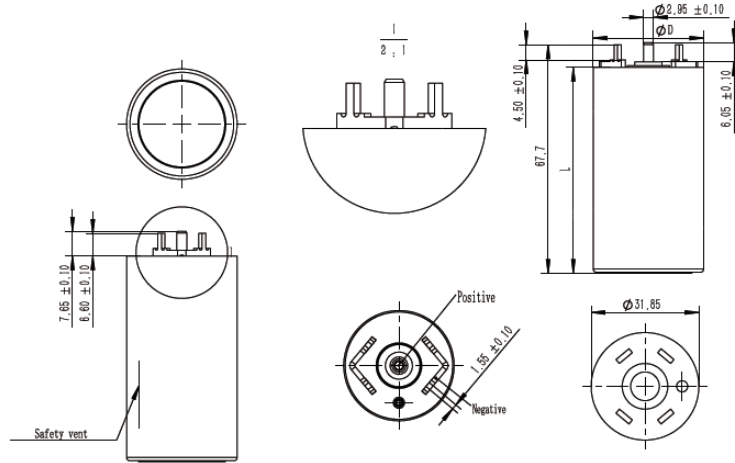
Service Lifetime

Endurance	Product held at rated voltage in 65°C environment for 1500 hours	
	Change in capacitance (% drop from rated)	≤20%
	Change in ESR (% increase from maximum initial)	≤100%
DC Life	Product held at rated voltage in 25°C environment	
	Projected Life	10+ years
	Change in capacitance (% drop from rated)	≤20%
Cycle Life	Change in ESR (% increase from maximum initial)	≤100%
	Cycling from rated voltage to 50% voltage under constant current in 25°C environment	
	Projected Life	1,000,000 cycles
	Change in capacitance (% drop from rated)	≤20%
Storage Life	Change in ESR (% increase from maximum initial)	≤100%
	Stored uncharged in original packaging in 25°C environment	
Life	Life	4 years

Physical Characteristics

Mechanical	Operation Vibration	IEC60068-2-6, SAE J2380
	Impact	IEC60068-2-27, SAE J2464

Outline Drawings:



Weight and Size:

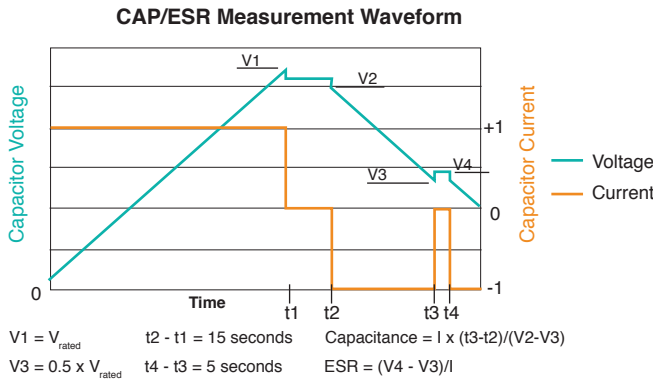
Weight: 62.9g | Size: L (Max.) 61.5mm D (Max.) 33.3mm

Naming Rules:

Type	Capacitance	Dash	Rated Voltage	Dash	Termination
SC Supercapacitor Cell	0350 = 350F	-	270 = 2.7V	-	RSS = Radial Square Solder

Notes:

1. Measure capacitance and DC internal resistance at 25°C under specified test current per Figure 1



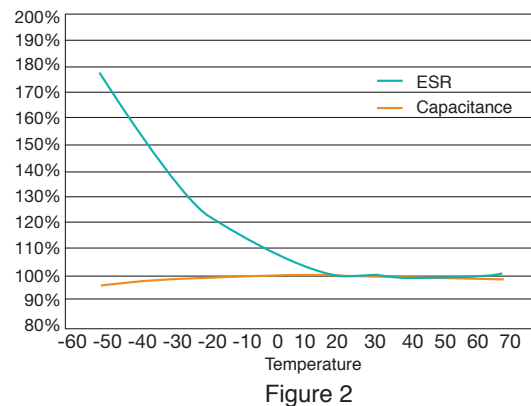
2. Surge voltage is non-repeatable and duration cannot exceed 1s
3. Corresponding current value after 72 hours of rated voltage at 25°C
4. $\Delta T = I_{rms}^2 \times ESR \times R_{ca}$
5. $0.5CV^2/3600$
6. $0.5C(V_{nom}^2 - V_{min}^2)/3600$

$$7. Wh_{usable} / \left(\frac{\pi r^2 (mm) \times L (mm)}{1 \times 10^6} \right)$$

$$8. Wh_{usable} / \text{weight (kg)}$$

$$9. \text{Per IEC62391-2 } P_d = \frac{0.12V^2}{ESR_{DC} \times \text{weight (kg)}}$$

10. Test after the sample has been maintained at -50°C for 16 hours and the temperature raised 10°C each time and maintained for 1 hour, then test the sample Figure 2



Specifications are subject to change without notice.